Fashion-MNIST

Loads Fashion-MNIST IDX image files, validates them, visualizes samples, and performs GPU-ready tensor testing.

In [ ]:
```
import torch
import numpy as np
import struct
import matplotlib.pyplot as plt
from pathlib import Path

device = "cuda" if torch.cuda.is_available() else "cpu"

print("CUDA:", torch.cuda.is_available())
print("Device:", device)

if torch.cuda.is_available():
    print(torch.cuda.get_device_name(0))
```

Set Dataset Path

In [ ]:
```
DATASET_PATH = "train-images-idx3-ubyte"

assert Path(DATASET_PATH).exists()

print(DATASET_PATH)
```

Load IDX Images

In [ ]:
```
def load_idx_images(path):
    with open(path, "rb") as f:
        magic, n_images, rows, cols = struct.unpack(">IIII", f.read(16))

        print("Magic:", magic)
        print("Images:", n_images)

        data = np.frombuffer(
            f.read(),
            dtype=np.uint8
        ).reshape(n_images, rows, cols)

    return data

images = load_idx_images(DATASET_PATH)

print(images.shape)
```

Dataset Statistics

In [ ]:
```
print("Min:", images.min())
print("Max:", images.max())
print("Mean:", images.mean())
print("Std:", images.std())
```

Visualize Samples

In [ ]:
```
idx = np.random.choice(
    len(images),
    25
)

fig, axes = plt.subplots(
    5,5,
    figsize=(8,8)
)

for ax, i in zip(axes.flatten(), idx):
    ax.imshow(images[i], cmap="gray")
    ax.axis("off")

plt.tight_layout()
plt.show()
```

Convert to GPU Tensors

In [ ]:
```
tensor_images = torch.tensor(
    images,
    dtype=torch.float32
) / 255.0

tensor_images = tensor_images.unsqueeze(1)

tensor_images = tensor_images.to(device)

print(tensor_images.shape)
print(tensor_images.device)
```

Batch Testing

In [ ]:
```
batch = tensor_images[:1024]

print(batch.shape)

memory_mb = (
    batch.element_size()
    * batch.nelement()
) / (1024**2)

print("Memory MB:", round(memory_mb,2))
```

Notes

Upload label files later if you want training support.